CSE 21 Winter 2026
Homework 2
Due date: Monday, October 12 at 11:59pm

INSTRUCTIONS

This homework should be done in **groups** of one to four students, without assistance from anyone besides the instructional staff and your group members. Homework must be submitted through Gradescope by a **single representative** of your group and received by **11:59pm** on the due date.

Students should consult their textbook, class notes, lecture slides, podcasts, group members, instructors, TAs, and tutors when they need help with homework. You may ask questions about the homework in office hours, but questions on Piazza should be private, visible only to instructors.
This assignment will be graded for not only the *correctness* of your answers, but on your ability to present your ideas clearly and logically. You should explain or justify, present clearly how you arrived at your conclusions and justify the correctness of your answers with mathematically sound reasoning (unless explicitly told not to). Whether you use formal proof techniques or write a more informal argument for why something is true, your answers should always be well-supported. Your goal should be to **convince the reader** that your results and methods are sound.
KEY CONCEPTS Inclusion/Exclusion, Binomial Identities, Stars and Bars.
(Note: For this homework, you can leave your answers in terms of exponentials, factorials, binomial coefficients, etc.)

(Note: for justifying counting arguments, a good rule of thumb is to explain how you came up with every term and factor of your answer. You can leave your answer in terms of exponentials rather than compute the exact numerical value.)

1. A license plate is a string over the alphabet of 36 characters consisting of 26 uppercase latin letters and 10 numerals (0 through 9) (*2 points for correct expression and 2 points for justification*)

   (a) (4 points) How many 7-character license plates start with a letter or end with a numeral?

   (b) (4 points) How many 7-character license plates use at least one A, at least one B and at least one numeral?

   (c) (4 points) How many 7-character license plates contain only numerals that all add up to 10?

   (d) (4 points) How many 7-character license plates are there that are all letters and they are in alphabetical order (repeats allowed)?

2. An art museum with 12 different rooms just got a shipment of 9 new (identical) prints (*2 points for correct expression and 2 points for justification*)

   (a) (4 points) How many ways can the museum curator arrange all 9 prints among the 12 rooms in the museum?

   (b) (4 points) How many ways can the museum curator arrange all 9 prints among the 12 rooms in the museum so that each print is in a different room?

   (c) (4 points) How many ways can the museum curator arrange all 9 prints among the 12 rooms in the museum so that at least 3 of the rooms are not empty?

   (d) (4 points) How many ways can the museum curator arrange all 9 prints among the 12 rooms such that at least one room has at least 3 prints?

(e) (4 points) How many ways can the museum curator arrange all 9 prints in 3 of the 12 rooms such that each room has exactly 3 prints?

3. (12 points) Compute the number of *integer solutions* for each equation:
(Include a justification)

(a)
$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 30$$
$1 \leqslant a_1$
$2 \leqslant a_2$
$3 \leqslant a_3$
$4 \leqslant a_4$
$5 \leqslant a_5$
$6 \leqslant a_6$

(b)
$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 30$$
At least one of the variables is less than or equal to 3.

(c)
$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 30$$
$0 \leqslant a_1 \leqslant 7$
$0 \leqslant a_2 \leqslant 7$
$0 \leqslant a_3 \leqslant 7$
$0 \leqslant a_4 \leqslant 7$
$0 \leqslant a_5 \leqslant 7$
$0 \leqslant a_6 \leqslant 7$

4. (a) (8 points) For $n \geqslant 0$ and $k \geqslant 0$ consider the identity:
$$k\binom{n}{k} = n\binom{n-1}{k-1}$$
Prove it combinatorially by counting the same set in two different ways.

(b) (8 points) For $n \geqslant 3$, prove that:
$$\binom{n+1}{3} = 1(n-1) + 2(n-2) + 3(n-3) + \cdots + (n-3)3 + (n-2)2 + (n-1)1$$
using a combinatorial argument.
[Hint: count the fixed density binary strings with 3 ones based on where is the middle one)]